

What's New and What's Next for Capella

THURSDAY, SEPT 10th, 2026



**Stéphane
LACRAMPE**
OBEO

Don't miss out the next events!



WEBINAR

with



Connect Capella to Your AI

Ecosystem by Stephane Lacrampe (Obeo)

Sept 24, 2026

Online

<https://mbse-capella.org/webinars.html>



Save the date! The event will take place December 1-3, 2026

Program announced on September 17

https://mbse-capella.org/capella_days.html

**ARCADIA AND
CAPELLA TRAINING**

Sept 28 - Oct 1 & 5-6, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Nov 16-23, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Dec 14-21, 2026

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<https://www.obeosoft.com/en/services/training/capella-training>





Stephane LACRAMPE
*Managing Director, North and
South America*
Obeo

Agenda

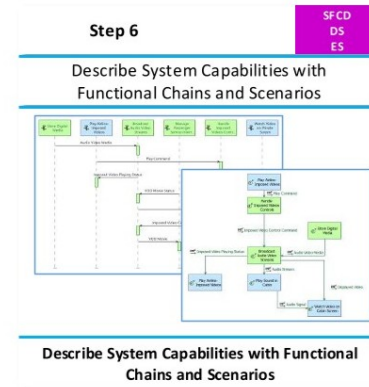
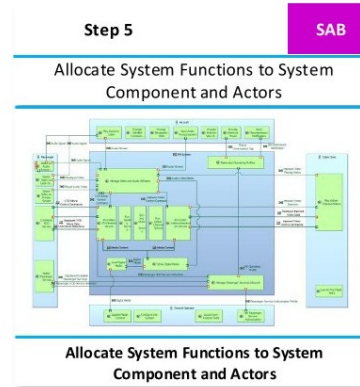
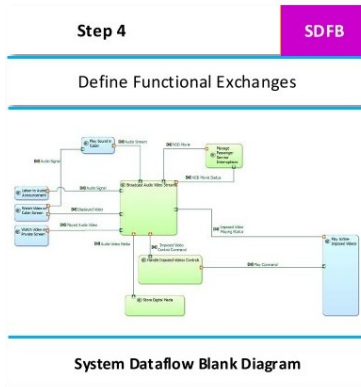
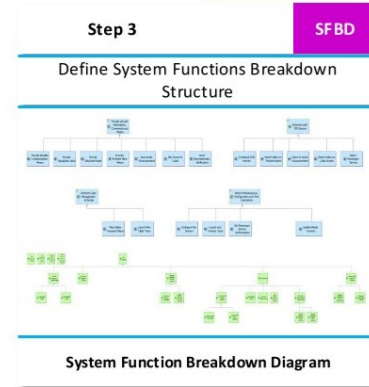
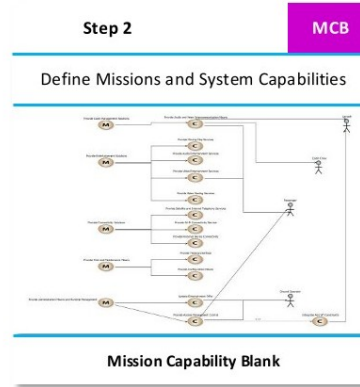
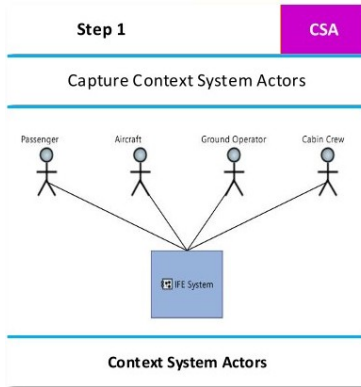
- **Introduction to Arcadia & Capella**
- The Capella Community
- What's New in Capella 7.1.0
- Capella Open-Source Add-ons
- Ecosystem Products
- Capella for SysON
- Q&A

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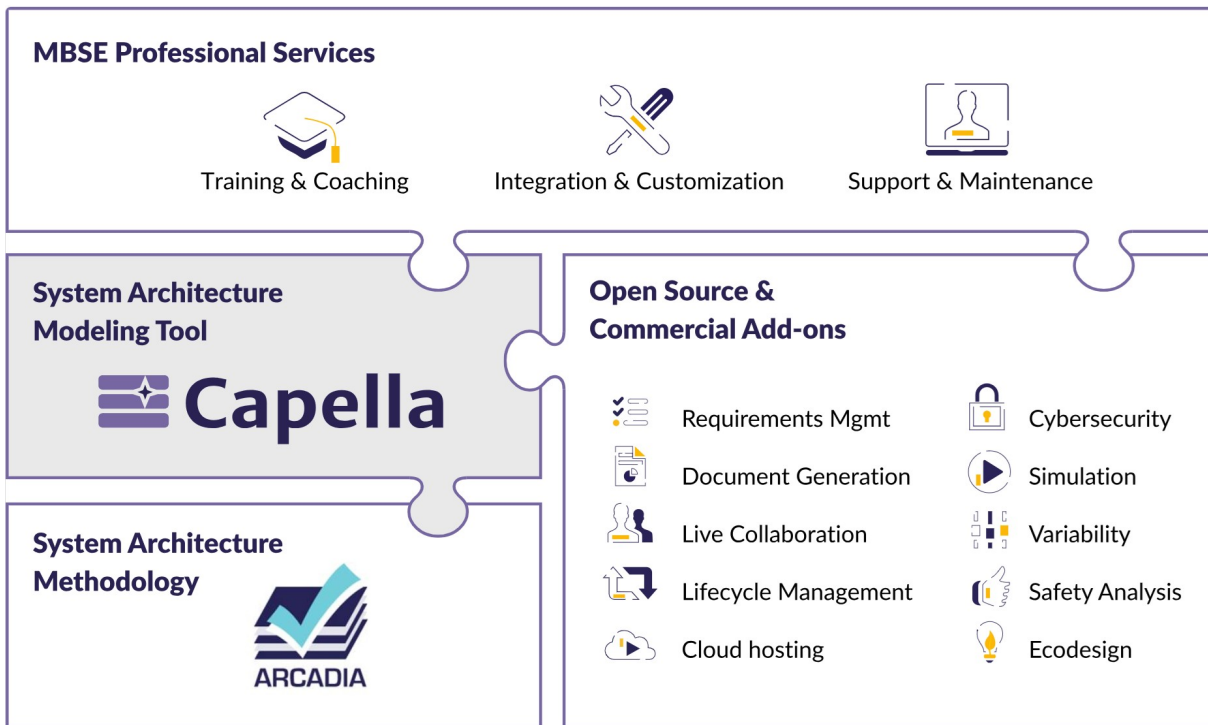
Physical Architecture
How the system will be developed and built



Arcadia/Capella Views



Capella ecosystem



Capella, the modeling tool implementing the Arcadia methodology, is the cornerstone of a comprehensive MBSE ecosystem.

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A Thriving Community, Built Over a Decade

- 80+ webinars organized since 2017
- 9 editions of Capella Days (annual online conference)
- 180+ videos on the Capella YouTube channel
- 500+ LinkedIn posts/year
- Active forum: ~150 messages/month



Capella website content available in 8 languages

WORKBENCH ▾

TOOL FEATURES

GET STARTED

ADD-ONS

 ÜBERSICHT

 APERÇU

 RESUMEN

 VISÃO GERAL

 概要

 개요

 概述

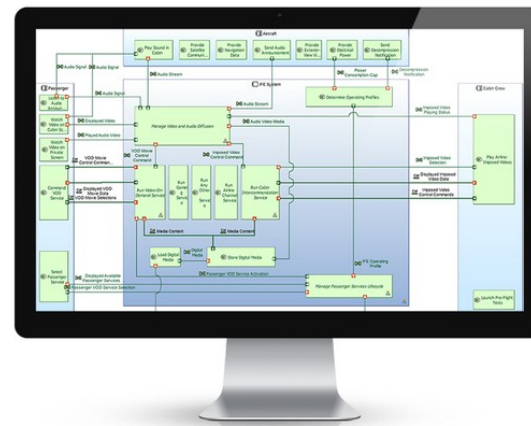
 ภาพรวม



Capella เป็นเครื่องมือ MBSE แบบโอเพนซอร์สที่สร้างแบบจำลองที่ให้แผนภาพหลากหลายประเภท โดยจัดเรียงตามมุมมองทางวิศวกรรม 5 มุมมองที่กำหนดไว้ใน Arcadia.

Capella มีฟังก์ชันการทำงานที่ช่วยให้สามารถปฏิบัติตามวิธีการของ Arcadia ได้อย่างราบรื่น เช่น การเข้าถึงกิจกรรมวิธีการที่มีบริการให้ใช้ทั้งหมด (Activity Explorer) และชุดเครื่องมือเร่งความเร็วในการสร้างแบบจำลองหรือเครื่องมือช่วยกระบวนการทางวิศวกรรมมากมาย ซึ่งช่วยอำนวยความสะดวกในการทำงานประจำวันของวิศวกรระบบ เพื่อให้พวกเขาสามารถมุ่งเน้นไปที่การออกแบบแทนที่จะต้องเสียเวลาไปกับการบำรุงรักษาแบบจำลอง

คุณสนใจของ Capella (ภาษาอังกฤษ) ➔



Capella at local and international conferences

- INCOSE International Workshop & International Symposium
- INCOSE AOSSEC, EMEA, GLRC, WSRC (US), SETE (Australia), Japan, Korea, India...
- ESA MBSE Conference, IEEE SysON, Siemens Realize Live, CSDM, AFIS...
- 160+ conferences since 2015
- Sponsoring, Capella booth, talks, tutorials...
- 1,500+ organizations using or having used Capella



2026 : a very active year

- 7 webinars since January
- Capella Days 2026 (Dec 1–3): Call for Proposals closed with strong response
 - talks from Rolls-Royce, Indra Space, Bluekei already preselected
- On the road: INCOSE IS (Yokohama), EMEASEC (Sweden), Dutch SE Days, TdSE, CSD&M, ASEC (UK), AeroSEC (India)...
- Growing local footprint: Capella meetups (Australia, Brazil...)

**WEBINAR**

Save the date
**Practice in COMAC to Conduct MBSA
in Avionics System Based on Capella**
中国商飞航电系统基于Capella的MBSA
应用实践

Tuesday, May 12th, 2026
4 PM-5 PM CST (Beijing)

**Hu
PO
COMAC****Renfei
XU
PGM**

Organized by
**OBEQ**

With the participation of
**COMAC PGM**



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Capella 7.1.0: Stronger foundations



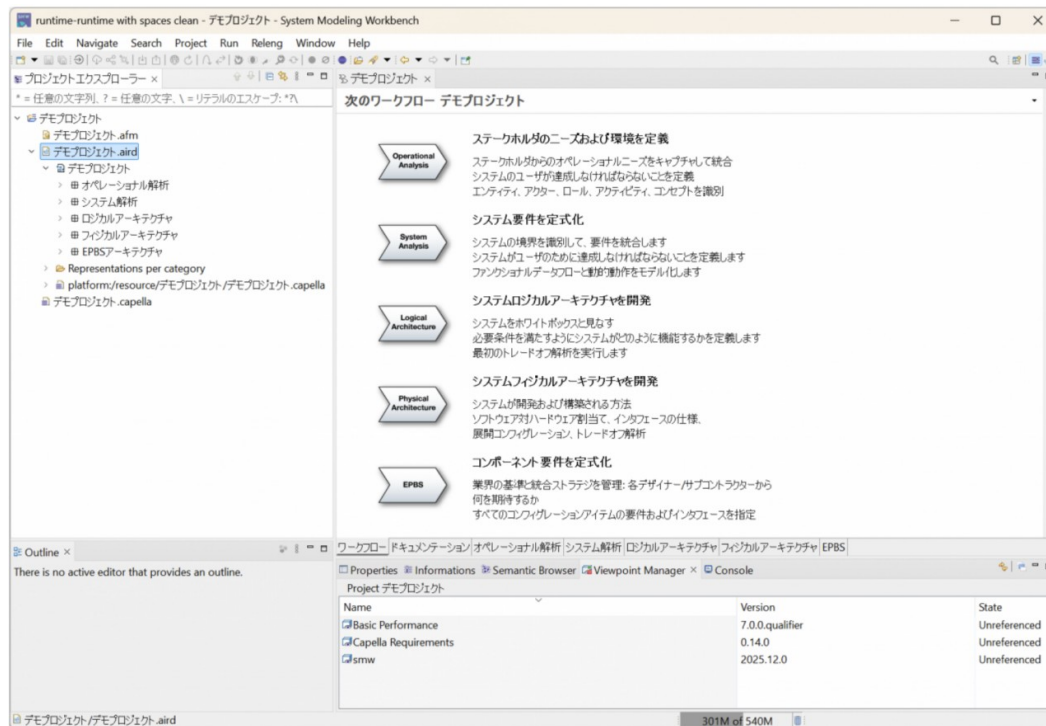
Building the future of Capella starts with strengthening its foundations: a more robust, secure and easier to maintain platform over the long term.

- **From Java 17 to Java 21 (more recent and sustainable version).**
- **CyberSecurity focus: nine CVEs addressed through dependency updates and removals, including:**
 - Removal of Mylyn features
 - Migration to reload4j 1.2.26
 - Migration to Guava 33
 - Migration to SLF4J API and Logback 1.5.32
 - Migration to EGit/JGit 6.6.1



Capella 7.1.0: Internationalization

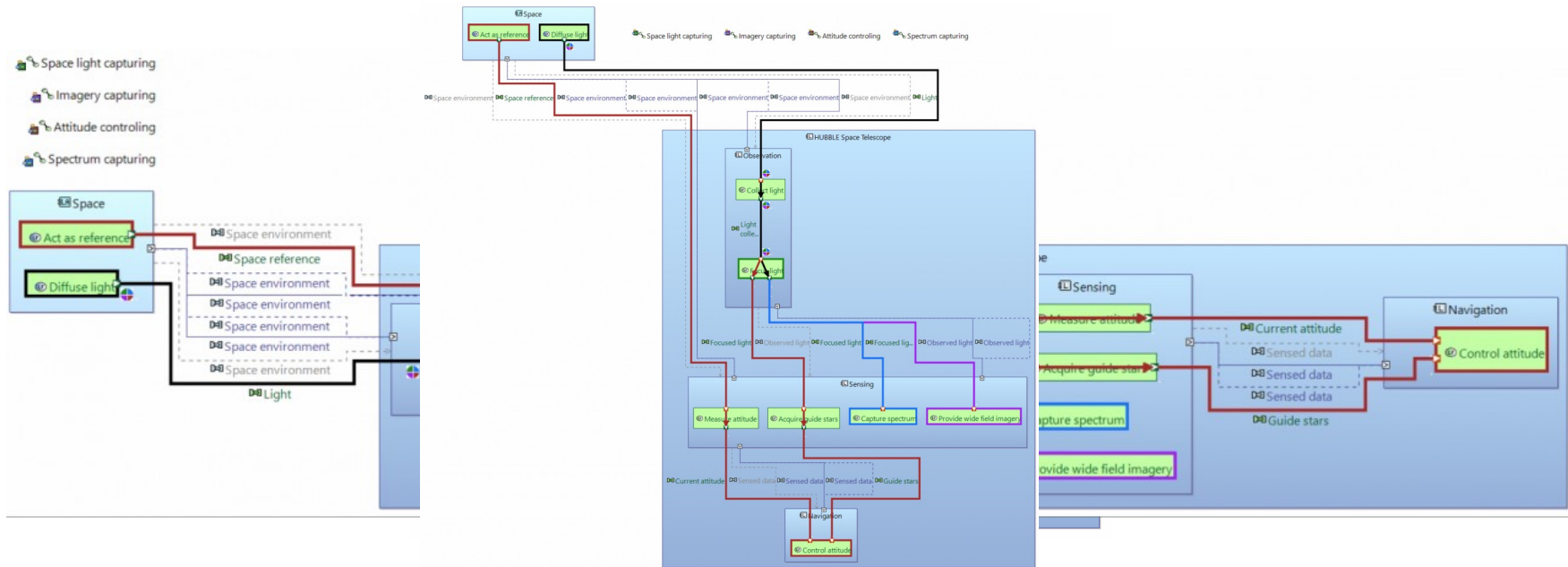
- The platform now provides the foundations needed to translate Capella's user interface into different languages.



Capella does not ship with translations out of the box yet: translations need to be provided separately.

Capella 7.1.0: Enhanced Layouting

- Diagram Layouts: probably the number one pain point in Capella
- Capella 7.1.0 includes a first integration of the Eclipse Layout Kernel (ELK) add-on, giving you access to a variety of automatic layout algorithms (tested on xAB diagrams)



Capella 7.1.0: Other improvements

- **Other improvements**
 - Project migration through APIs
 - REC/RPL fixes
 - Color disposal issues on Linux



Obeo commitment to Capella



- Obeo has been present **from the start**, supporting **Thales** in developing Capella.
- Our role has expanded since Capella was put in **open-source** in 2015
 - Actively supported the **international development** and **adoption** of Arcadia and Capella, establishing it as **leading reference** in the MBSE landscape
 - Actively maintained **key technologies** that support Capella, including Eclipse Sirius, EMF Services, GMF Runtime, and other Eclipse technologies
- This Capella 7.1.0 is the beginning of an **extended involvement** of Obeo in Capella
 - **New release cycle** (2026.X, 2027.X...) with 3 to 4 releases per year
 - New committers and project **co-leadership**
 - Extended direct **investments** and **contributions** to Capella
 - And other exciting topics...

What's Next for Capella 2026.11.0 ?

- We are ramping up our capability to release Capella and all the underlying components
- Significant Eclipse Platform upgrade (from 2023.03 to 2025.09)
 - Upgrade on all Capella dependencies
 - Improved OS compatibility and decreased CVE exposure
- Activation of the new version numbering (CalVer)
- Various bug fixes and improvements
- ELK integration and possible improvements



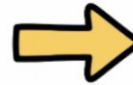
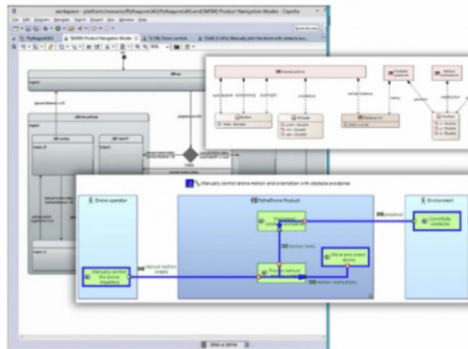
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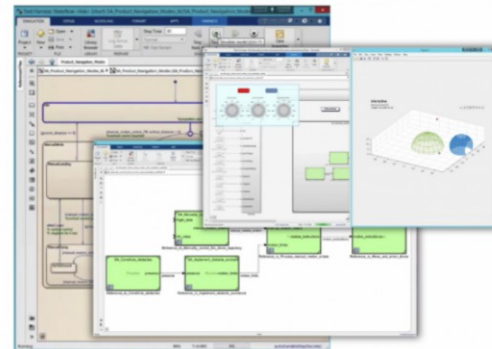
Capella Simulink® Connector

- **New** Open-Source add-on
 - **Developed by Thales** and open-sourced by Obeo
<https://github.com/eclipse-capella/capella-simulink-connector/releases/tag/0.2.0>
 - Integrate Capella state machines and functional chains with Simulink® for simulation
- Upcoming: Installation documentation and webinar presentation

 **Capella**

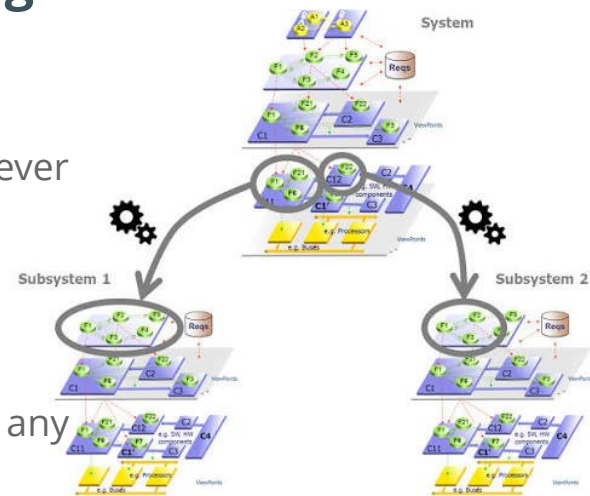


SIMULINK®



System to SubSystem transition: Safer Perimeter changes

- **Before**
 - Transition **assumed** the subsystem's component list (its "perimeter") never changed
 - Changing it often led to **model corruption**
- **Now**
 - Component selection is reviewed in the **transformation dialog** before any change is applied
 - Elements are **matched by role**, not technical ID → existing elements are updated, not destroyed and recreated
 - Impact on other model elements and diagrams is minimized
 - **Cancel button fixed**: no longer triggers unwanted modifications



Available in the latest version of the add-on and funded by Rolls-Royce



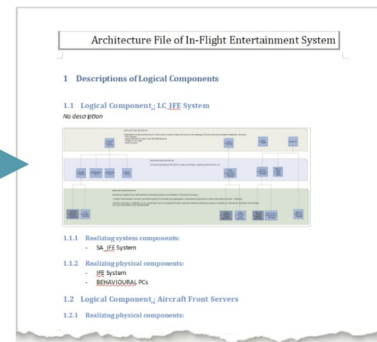
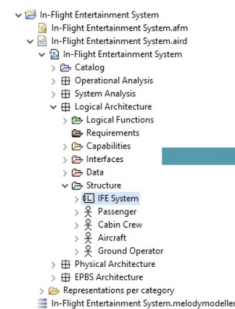
M2Doc: Keeping Word Generation Up to Date

- Recent evolutions

- Office 365 support** & plain-text instructions (since M2Doc 4.0.0) ->better compatibility, easier to inspect templates
- Continuous improvements** to HTML-to-Word integration (lists, tables, images)
- Section & page numbering via bookmark references for **accurate cross-references** in generated documents
- New: call a Semantic Browser query directly from a template : ***myLogicalFunction.getSBQuery('Functional Exchanges')***

- Looking ahead

- M2Doc 5.x is based on AQL 8 — adoption tied to Capella's own move to AQL 8
- Roadmap mainly driven by **customer needs**



Python for Capella — Scripting Made More Powerful

- **Recent evolutions (1.4.0)**
 - New Diagram API: create diagrams, generate mappings, customize styles, by script
→ Demo
 - Broader API coverage: fixes across Capella core, PVMT and Requirements add-ons
 - Better packaging: core API and sample scripts now separate, more portable
 - Python 3.12 compatibility
- **Looking ahead**
 - Acceleo-based generator opens the door to auto-generating APIs for additional Capella add-ons

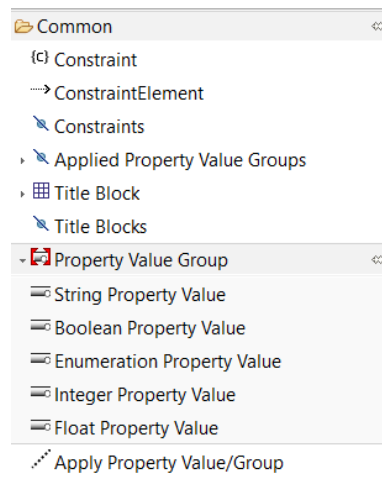
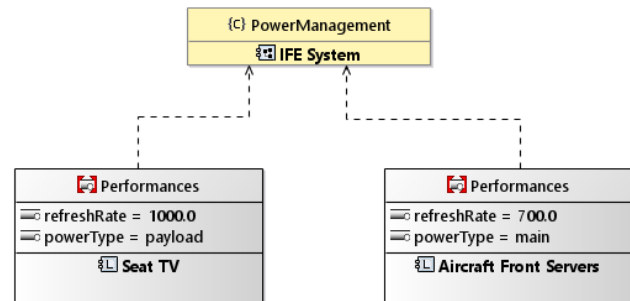
Capella Traceable Properties Values Edition

• What it does

- Makes non-functional requirements (mass, cost,...) easier to create, apply, visualize and trace in Capella
- Built on the Holistic Requirements Model: functional requirements say what the system does, non-functional requirements constrain how well

• How it works

- No new metamodel elements : reuses existing Capella concepts: Property Values, Property Value Groups, Applied Property Value Groups, Generic Trace links
- Adds usability: new palette tools + new diagram types on top of these existing concepts
- Property Value support directly on existing Capella diagrams + dedicated traceability diagrams for non-functional requirements
- Open-source (EPL), Available by end of November



Non-Functional Requirement Creation & Traceability using Property Values



Joshua Wedgwood
Rolls-Royce



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Team for Capella



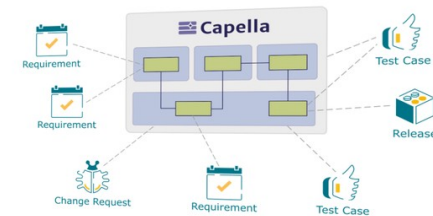
- **From 6.1.0 to 7.0.1_202601**

- **Authentication overhaul:** OpenID Connect rework, Authorization Code + PKCE, BCrypt-encrypted credentials, secrets via environment variables
- **Semantic Browser performance:** refresh time cut from ~1 min to **under 1 sec** on large connected projects
- **Server robustness:** CDO revision cache memory leak fixed, continuous Jetty/Jenkins CVE cleanup
- More reliable Import/Export wizards (special characters, **library detection**, unshared projects)

- **7.1.0 (latest) : Secure by Default**

- WSS/HTTPS now the **default connection mode**
- New: **per-repository capabilities** : control which Capella add-ons are authorized, per project
- **Extended OIDC:** opaque access token support (introspection)
- Aligned with Capella 7.1.0 foundations: Java 21, updated dependencies

Publication for Capella



- **Recent (2026.5 – 2026.7)**

- Broader ALM coverage: Jama Connect, Jira Cloud (via SodusWillert)
- New Organizations & Server Admin role for scalable access management
- Foundations for version control: history, baselines, previews
- Diagram images now published as Polarion work items

- **In progress**

- Bringing Capella–CodeBeamer integration to full traceability feature set
- Decorators in the online explorer (e.g., surfacing Filtering add-on filters)
- API access to object history, easing integration into customers' software landscape

- **Looking ahead**

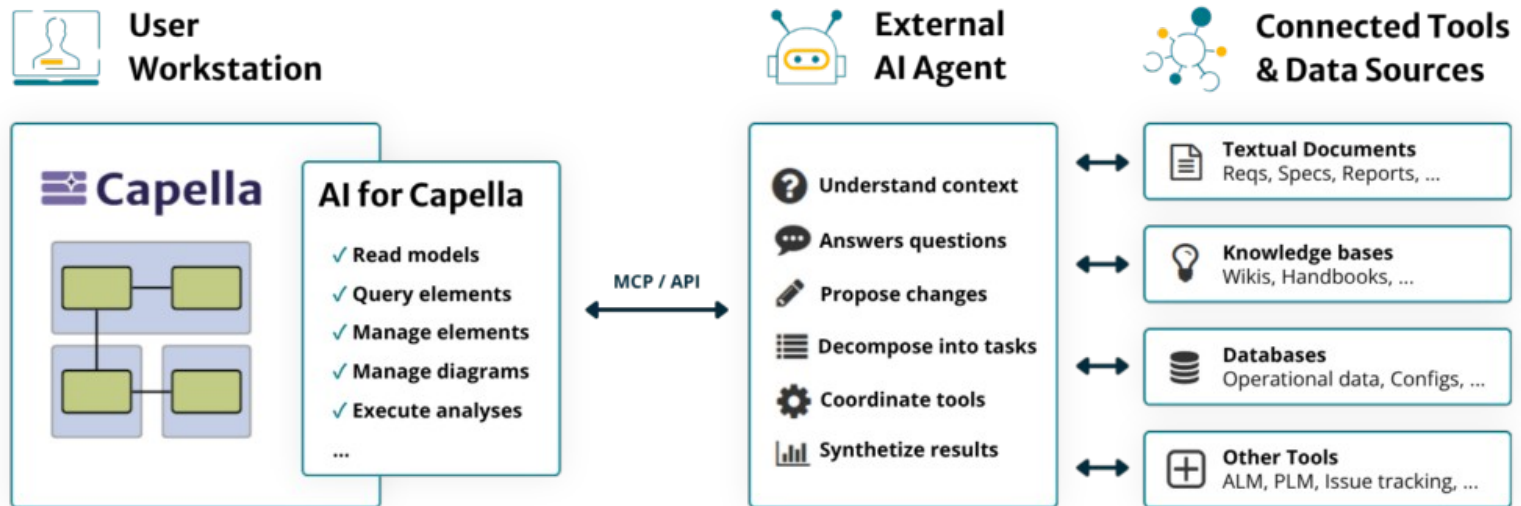
- AI-powered natural language querying of models
- Discussions on model elements

System Modeling Workbench (SMW)

- **2606**
 - Now based on Capella 7.1.0, requires Java 21
 - Key fix: **concurrency issue** in "update" that could corrupt models
 - Focus: **security & robustness**
- **What's next (2612)**
 - Eclipse 2025.09 upgrade & removal of legacy Chromium browser (both **CVE-driven**)
 - Exploring: rich-text images, SMW subsystem reuse from Teamcenter
 - New: multiple language support (installer-configurable), building on Capella 7.1's i18n foundations
 - Packaging the **EDA VSE add-on**

AI for Capella: Capella as an Agent-Ready Platform

- Rather than embedding one built-in assistant, AI for Capella **connects Capella to your AI ecosystem** : Claude, Codex, Mistral, a local model, or any agent platform
- **Controlled**, traceable interactions : **engineers stay in the loop**
- **Enterprise-ready** add-on for Capella Desktop



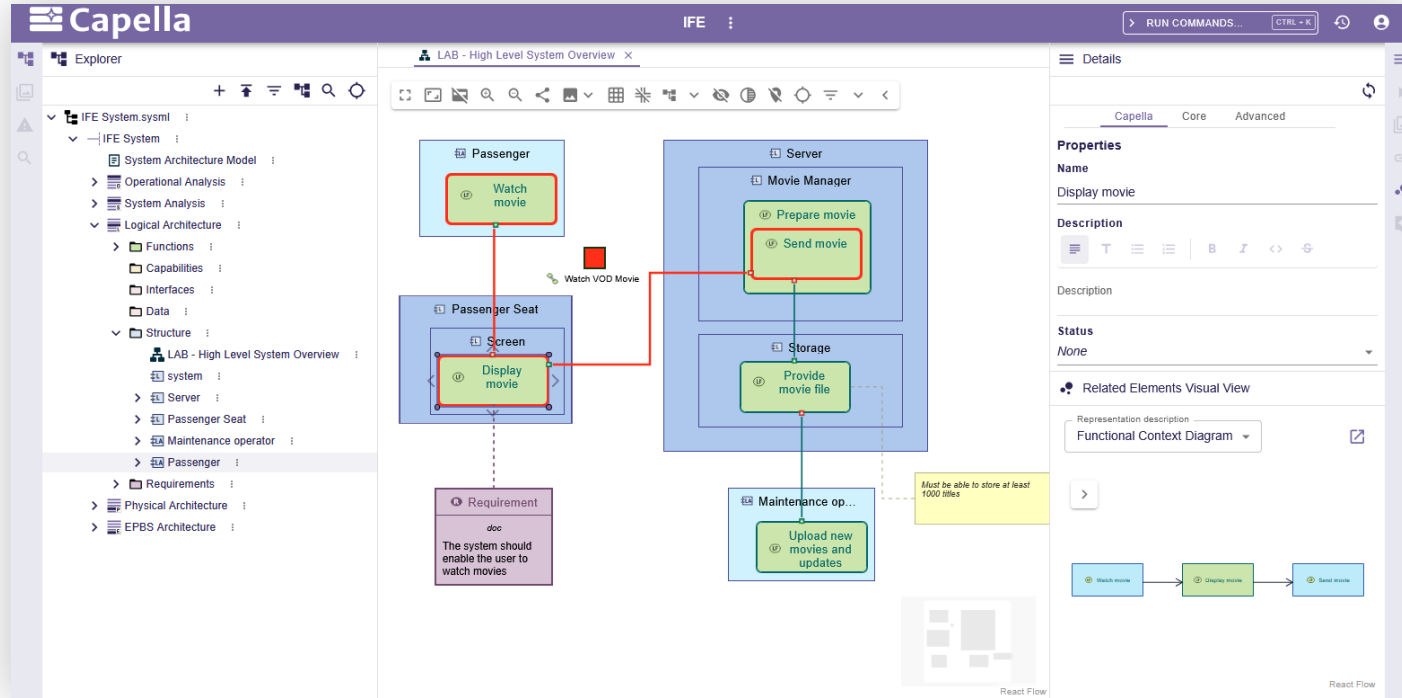
Other add-ons



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Capella for SysON: Towards the Next Generation of Capella



A new open-source initiative exploring the convergence of Capella, Arcadia, SysML v2, and web-based modeling

Introducing SysML v2

- Obeo is an **OMG member**
 - Part of the **SysML v2 Submission Team (SST)**, and actively contributes to the definition of SysML v2
- **Key evolutions / advantages**
 - New OMG standard, formally approved in **2025**
 - Built on KerML foundations: improved precision, expressiveness, and consistency
 - Native **textual** + **graphical** notation
 - Standardized API : positioning SysML v2 as the future **interoperability** standard across MBSE tools
- **The challenge**
 - SysML v2 is a **rich, powerful**, and **complex** general-purpose language
 - Its **level of abstraction** is **unchanged** from v1: it tells you how to express a model, not how to structure your systems engineering approach



**A structured method like Arcadia remains just as relevant,
and just as necessary, on top of SysML v2**

Sirius Web and SysON



- **Sirius Web**

- Cloud/low-code evolution of Eclipse Sirius: a low-code platform for building graphical modeling tools
- Modeling directly in the browser, built on modern web tech (React, Spring, GraphQL...)

- **SysON**

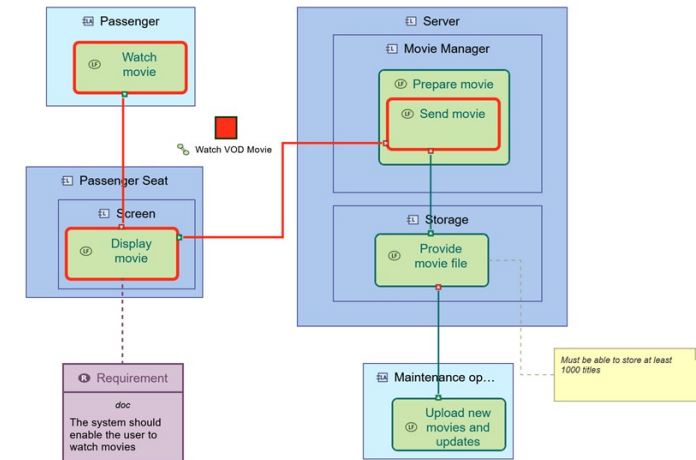


- Leading open-source, web-based SysML v2 editor, built on Sirius Web
- Co-developed by Obeo & CEA List, hosted at the Eclipse Foundation
- Flexible by design: customizable for domain-specific methodologies, without forking
- Already proven on real methodology extensions

**This is precisely what makes it possible to bring Arcadia
into a SysML v2, web-based environment**

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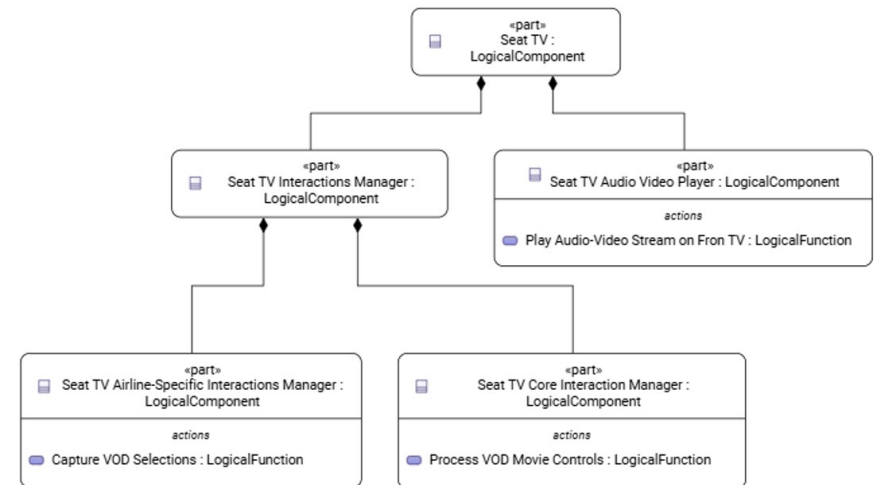


The Arcadia SysML v2 Library

- **The approach**

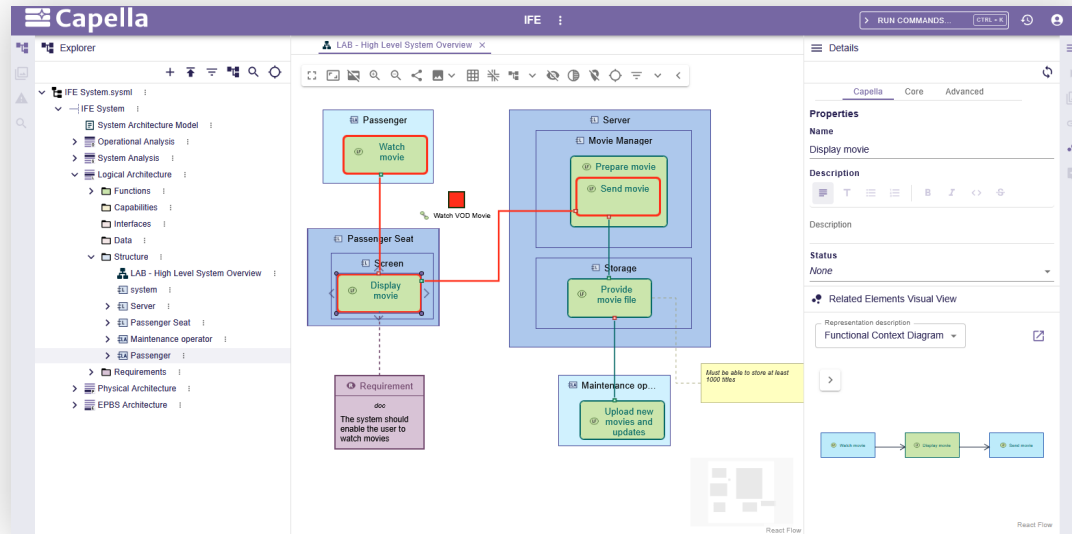
- Priority #1: **interoperability**. Models should stay as close as possible to "natural" SysML v2.
- A **minimal, essential semantic layer**: only the core concepts that matter (Functions, Components, Exchanges...)
- **Generic concepts** (Function, Component, Port) rather than perspective-specific ones (Logical/Physical Function): easier tooling, easier reuse across perspectives
- Based on **15+ years of Arcadia/Capella usage**: focus on what the community actually uses, not the full technical metamodel

```
part def Component :> ArcadiaElement {  
  private doc /* A generic Arcadia component, used as  
  attribute isActor : Boolean default false;  
  attribute isHuman : Boolean default false;  
  alias allocatedFunctions for performedActions;  
  alias subComponents for subparts;  
}  
  
port def ComponentPort :> ArcadiaElement {  
  private doc /* A generic Arcadia Component Port,  
  attribute portDirection : FeatureDirectionKind;  
  ref item allocatedFunctionPorts: ExchangeItem [*];  
}
```



Capella for SysON: A First Look

- What it brings
 - Capella-inspired **graphical editors**,
 - built on top of SysML v2 elements specialized with the Arcadia library
 - Same **modeling experience** and vocabulary Capella users already know, in a **browser**



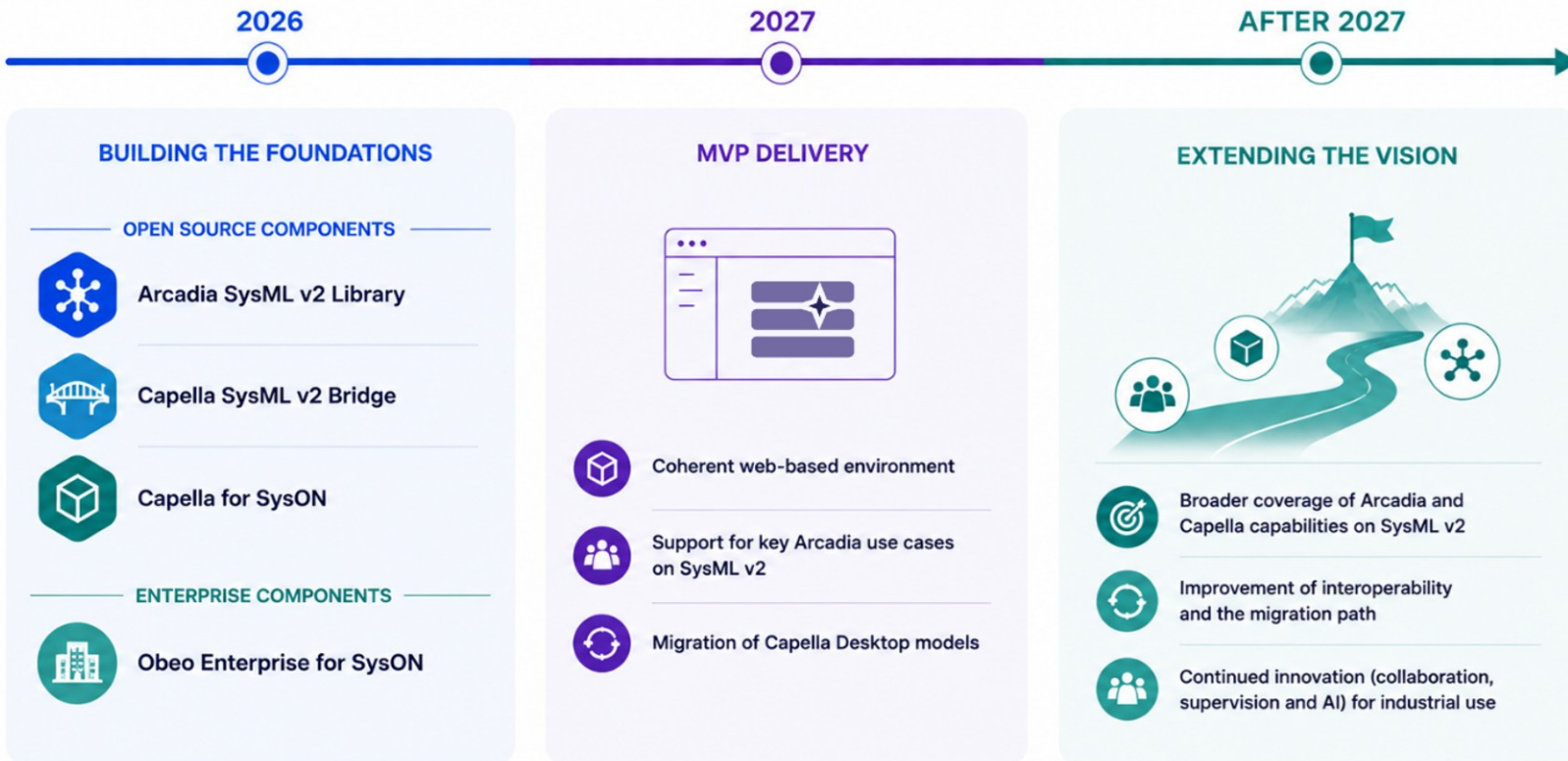
The Ambition: Building the Next Generation of Capella

- **A credible, progressive path toward a Capella environment :**
 - **Web-based :** easier deployment and collaboration in modern engineering organizations
 - **Compatible with SysML v2 :** connecting Arcadia-based engineering to the emerging standard and its ecosystem
 - **Built for collaboration :** leveraging SysON's architecture: REST APIs, textual syntax
 - **Grounded in Arcadia :** preserving the methodological approach that is central to Capella
- **But not an overnight replacement**
 - **Capella is mature:** hundreds of wizards, helpers, and model transformations refined over 15+ years
 - Building the next generation means **progressively** bringing that value to a new foundation, not just reproducing diagrams in a browser

 <https://github.com/eclipse-capella/capella-for-syson>

Roadmap

Towards the Next Generation of Capella



The Capella SysML v2 Bridge: A Path Forward, Today

- **What it does**

- Exposes **existing Capella Desktop models** in SysML v2 : no need to leave Capella Desktop
- SysML v2 closed many gaps of v1 by moving **closer to Arcadia's concepts** : the transformation is far more natural than a v1 transformation would have been

- **Two use cases**

- A **migration path** toward SysML v2 / Capella for SysON
- Immediate **interoperability**: connect today's Capella models to the SysML v2 ecosystem (simulation, analysis...), even before Capella for SysON is mature

 <https://github.com/eclipse-capella/capella-sysmlv2-bridge>

Keep Modeling, and Help Us Shape and Build What's Next

- **If you use Capella today, keep using Capella**
 - Capella Desktop remains the **mature solution** for **operational** Arcadia projects : Capella for SysON is still **early-stage**
 - The path forward, in two steps: (1) expose Capella models in SysML v2 today via the Bridge, (2) migrate to Capella for SysON as it matures
- **Join the initiative**
 - Share your use cases, experiment, give **feedback**
 - **Co-fund** specific developments if this is **strategic for your organization**

**The future is not Capella or SysML v2.
It's Capella, with SysML v2.**

<https://blog.obeosoft.com/capella-for-syson>

Wrap-up

- Capella 7.1.0 is **stronger**, more **secure**, and more international — and it's just the beginning of a **bigger commitment**
- Obeo is investing **more**: 3 to 4 releases a year, new committers, extended contributions
- The ecosystem is thriving: new add-ons, real industrial contributions , a growing global community
- Capella for SysON: our path to the future



Thanks for listening!

Do you have any questions?

Let's keep on the discussion on the forum after the webinar:

<https://forum.mbse-capella.org/>

Don't miss out the next events!



WEBINAR

with



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Ecosystem by Stephane Lacrampe (Obeo)

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<https://www.obeosoft.com/en/services/training/capella-training>



Stay Tuned!



mbse-capella.org/capella_days_2026.html

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